

PETRUN', Russian Federation

WMO: 233240

Lat: **66.4711N**Lon: **60.7544E**Elev: **203**StdP: **14.59**Time Zone: **3.00 (E03)**Period: **09-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.6	-35.5	-45.2	0.4	-40.2	-40.3	0.5	-35.3	20.2	10.6	18.3	13.4	4.4	20	0.754

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	81.4	65.2	77.3	63.2	72.5	60.8	66.8	78.7	64.6	74.7	62.2	70.6	6.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.4	85.1	70.4	60.2	78.4	69.1	58.0	72.3	67.6	31.5	78.1	29.8	74.9	28.0	70.6	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
17.2	15.6	14.0	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-41.6	85.0	7.1	3.9	-46.7	87.8	-50.8	90.0	-54.8	92.2	-60.0	95.1
			-41.4	69.4	6.5	2.4	-46.0	71.1	-49.8	72.5	-53.4	73.9	-58.2	75.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.9	-4.2	0.8	11.9	24.0	35.3	51.1	58.8	51.5	43.3	29.5	12.8	6.3
	DBStd	24.15	17.14	17.14	14.76	10.46	10.52	9.53	8.59	7.30	6.87	9.43	15.32	18.13
	HDD50	9008	1680	1378	1180	779	478	99	14	70	221	636	1117	1356
	HDD65	13963	2145	1798	1645	1229	922	427	235	423	650	1101	1567	1821
	CDD50	580	0	0	0	0	21	133	288	116	22	0	0	0
	CDD65	58	0	0	0	0	0	11	43	4	0	0	0	0
	CDH74	701	0	0	0	0	6	123	533	39	0	0	0	0
	CDH80	148	0	0	0	0	0	17	128	3	0	0	0	0
Wind	WSAvg	7.1	6.6	6.9	7.4	7.8	7.7	7.4	6.3	6.6	6.7	7.1	7.4	7.4
Precipitation	PrecAvg	19.2	1.1	1.0	1.0	1.0	1.5	1.7	2.3	2.6	2.3	2.0	1.5	1.3
	PrecMax	27.1	2.1	1.7	2.2	2.3	2.8	3.4	4.8	5.0	4.4	4.1	2.6	2.3
	PrecMin	13.6	0.4	0.1	0.2	0.4	0.5	0.6	0.4	0.7	0.0	0.8	0.7	0.4
	PrecStd	3.5	0.4	0.4	0.4	0.5	0.5	0.7	1.2	1.1	1.0	0.7	0.5	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.8	32.6	38.6	49.0	74.3	82.6	86.6	79.5	69.0	46.8	36.5	33.6
		MCWB	30.4	31.0	33.2	40.6	56.9	64.4	66.6	66.1	59.0	43.7	36.0	32.6
	2%	DB	27.6	30.0	35.4	44.1	67.6	77.9	83.9	72.8	61.9	44.0	33.7	32.0
		MCWB	26.4	28.0	32.0	37.4	54.6	62.1	66.4	62.3	56.0	41.9	32.6	31.0
	5%	DB	23.0	27.3	33.5	40.2	59.6	73.4	80.3	68.8	57.5	41.9	32.5	30.1
		MCWB	22.1	25.8	31.2	35.6	50.1	60.4	64.9	59.8	52.7	40.2	31.7	28.8
	10%	DB	18.7	23.5	31.2	37.7	51.9	68.6	76.3	64.5	53.8	39.9	30.5	27.7
		MCWB	17.5	21.9	29.3	34.2	44.7	58.1	63.0	57.8	50.3	38.5	29.6	26.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.7	31.7	34.2	40.8	58.5	66.1	69.1	68.1	59.9	45.1	36.0	33.0
		MCDB	32.0	32.6	37.2	47.8	71.3	77.2	80.0	74.6	66.1	46.5	36.1	33.2
	2%	WB	26.5	28.5	32.8	38.3	54.7	63.9	67.6	63.9	57.1	42.2	33.1	31.2
		MCDB	27.6	29.8	34.8	42.9	66.0	74.7	80.5	70.2	61.0	43.4	33.5	31.9
	5%	WB	21.9	25.8	31.5	36.3	50.1	61.2	66.1	61.1	53.0	40.3	31.8	28.9
		MCDB	23.1	27.3	33.4	39.3	58.8	71.1	78.6	66.7	56.0	41.6	32.5	30.1
	10%	WB	17.4	22.1	29.3	34.6	45.3	58.7	64.2	58.5	50.5	38.6	29.7	26.5
		MCDB	18.5	23.7	31.3	37.2	51.5	68.3	74.3	63.3	53.3	39.8	30.5	27.6
Mean Daily Temperature Range	5% DB	MDBR	14.1	14.0	16.7	16.0	14.8	16.5	19.2	14.9	11.9	7.7	12.4	11.5
		MCDBR	11.9	11.0	11.1	15.8	26.3	23.7	25.7	22.6	17.6	8.1	7.5	8.8
		MCWBR	11.5	10.3	9.1	10.9	14.9	11.8	11.4	12.8	11.4	6.5	7.8	8.8
	5% WB	MCDBR	11.4	10.8	9.2	13.2	25.6	22.2	24.0	19.2	15.5	7.7	7.2	8.8
		MCWBR	11.2	10.3	8.1	9.6	15.0	12.3	11.9	12.4	10.4	6.5	7.6	8.8
Clear-Sky Solar Irradiance	taub	0.212	0.268	0.308	0.336	0.349	0.359	0.391	0.367	0.325	0.281	0.207	0.219	
	taud	1.886	2.019	2.056	2.084	2.248	2.438	2.352	2.396	2.471	2.244	1.923	1.899	
	Ebn at Noon	55	175	230	257	268	270	254	249	235	175	53	3	
	Edh at Noon	5	18	30	38	36	31	33	28	21	15	5	1	
All-Sky Solar Radiation	RadAvg	20	186	659	1235	1556	1581	1569	978	561	180	40	2	
	RadStd	2	30	75	103	169	145	230	107	59	29	5	0	

Historical Trends

		DBAvg
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